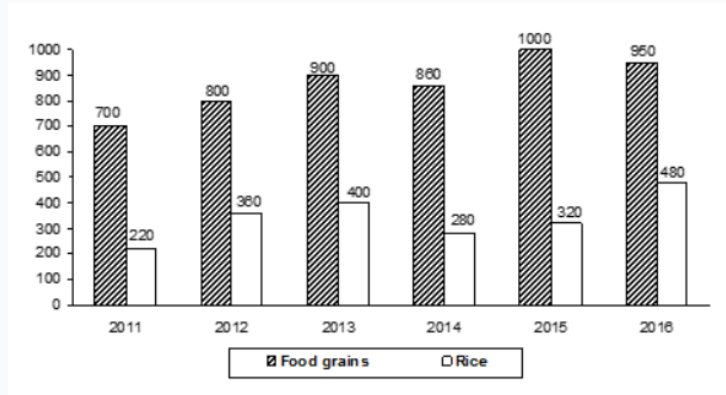


BAR GRAPH- DI 9-9-26

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The bar chart shows the total food grain and rice production in a country from 2011 to 2016 (in million tonnes).



Directions for questions 1 and 2: Select the correct alternative from the given choices.

In the year 2014, rice produced is what percentage of the total food grain production in that year?

- ☒ a) 32.55% (Your answer - ✓)
- ☐ b) 28.96%
- ☐ c) 35.93%
- ☐ d) 38.01%

Solution:

$$\text{Required percentage} : \frac{280}{860} \times 100 = 32.55\%$$

Choice (A)

Directions for questions 1 and 2: Select the correct alternative from the given choices.

The total rice produced during this period i.e., 2011–2016 is approximately what percentage of the total food grain production in the year 2013?

- ☐ a) 75.5%
- ☐ b) 229% (Correct Answer - ✓)
- ☐ c) 209%
- ☐ d) 133%

Solution:

Total rice produced = 2060 MT

$$\text{Required percentage} = \frac{2060}{900} \times 100 = 228.8\% \quad \text{Choice (B)}$$

Directions for questions 3 to 5: Type in your answer in the input box provided below the question.

In the year 2014, what is the quantity of food grains, produced other than rice (in million tonnes)?

Your Answer: 500 ✖

Correct Answer: 580 ✔

Solution:

Other food grains produced in 2014 = 860 – 280 = 580 MT. Ans : (580)

Directions for questions 3 to 5: Type in your answer in the input box provided below the question.

The production of food grains other than rice, as a percentage of the total food grain production, is the highest in which year?

Your Answer: 2015 ✖

Correct Answer: 2011 ✔

Solution:

The required percentages in the different years are

2015 = 68%

2014 = 67.4%

2013 = $\frac{500}{900} = 55.5\%$

2012 = 56%

2011 = 68.57%

Ans : (2011)

Directions for questions 3 to 5: Type in your answer in the input box provided below the question.

What is the average annual rice production from 2012 to 2016 (in million tonnes)?

Your Answer: 368 ✓

Solution:

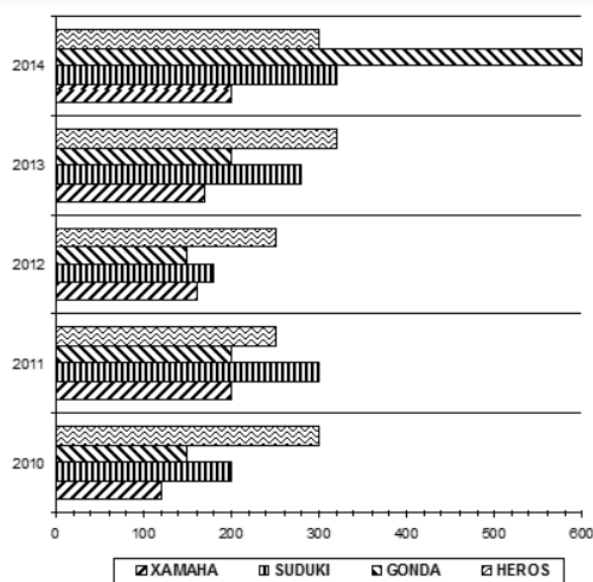
Total rice produced from 2012 to 2016 = 1840 MT

$$\Rightarrow \text{Avg} = \frac{1840}{5} = 368 \text{ MT.}$$

Ans : (368)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The bar graph gives the number of bikes sold in a city by different companies across five years. (Assume that there are no other companies which sell bikes in the city)



In which year did the company Xamaha record the highest percentage of total bikes sold by all the given companies in that year?

☒ a) 2010 (Your answer - ✖)

☐ b) 2011

☐ c) 2012 (Correct Answer - ✔)

☐ d) 2013

Solution:

$$\begin{aligned}\text{In 2010} &\Rightarrow \frac{120}{120+200+150+300} \\ &= \frac{120}{770} = \frac{12}{77} < 20\%\end{aligned}$$

$$\begin{aligned}\text{In 2011} &\Rightarrow \frac{200}{200+300+200+250} \\ &= \frac{20}{95} > 20\%\end{aligned}$$

$$\text{In 2012} \Rightarrow \frac{160}{740} = \frac{8}{37} > 20\%$$

$$\text{In 2013} \Rightarrow \frac{170}{970} < 20\%$$

$$\text{In 2014} \Rightarrow \frac{200}{1420} < 20\%$$

It is higher in 2012 as compared to 2011.

Choice (C)

The ratio of the total number of Suduki bikes sold during the years 2011, 2013, and 2014 to that of Heros bikes sold during the years 2010 to 2012 is same as that of the total number of Suduki bikes sold in 2008 to the total number of Heros bikes sold in 2008. If in 2008 the number of Suduki bikes sold was 171, then find the number of Heros bikes sold in 2008.

- ☐ a) 152 (Correct Answer - ✓)
- ☐ b) 142
- ☐ c) 132
- ☐ d) 144

Solution:

$$\frac{\text{Total Number of Sudukibikes sold for the years 2011, 2013 and 2014}}{\text{Total number of Heros bikes sold from 2010 to 2012}} = \frac{300+280+320}{300+250+250} = \frac{9}{8}$$

$$\frac{9}{8} = \frac{171}{\text{number of heros bikes sold in 2008}}$$

$$\Rightarrow 152.$$

Choice (A)

What is the total number of Gonda bikes sold during the given period?

- ☐ a) 1420
- ☐ b) 1300 (Correct Answer - ✓)
- ☒ c) 1280 (Your answer - ✗)
- ☐ d) 850

Solution:

$$\text{Total number of Gonda bikes sold} = 150 + 200 + 150 + 200 + 600 = 1300.$$

Choice (B)

Which of the following can be concluded?

- ☐ a) During the given period, in exactly one year the sales of bikes of all the given companies increased over the previous year.
- ☐ b) The number of Gonda bikes sold in 2013 is $33\frac{1}{3}\%$ more than those sold in 2012.
- ☒ c) During the given period, the total number of Heros bikes sold is less than the total number of Suduki bikes sold. (Your answer - ✖)
- ☐ d) Both (A) and (B) (Correct Answer - ✔)

Solution:

Only in 2013, the sale of all the bikes increased over 2012.

∴ Statement 1 is true.

The number of Gonda bikes sold in 2013 is $33\frac{1}{3}\%$ more than that in 2012.

∴ Statement 2 is true.

The total number of Heros sold is more than the total number of Suduki bikes sold during the given period.

∴ Statement 3 is false.

Choice (D)

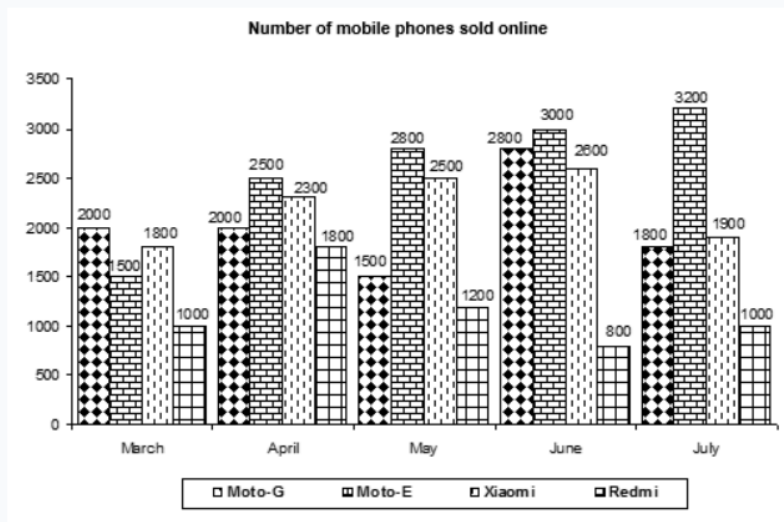
By what percentage did the number of Suduki bikes sold in 2013 increase as compared to those in 2012?

- ☐ a) $53\frac{4}{9}\%$
- ☐ b) $56\frac{1}{9}\%$
- ☐ c) 57
- ☒ d) None of these (Your answer - ✔)

Solution:

$$\begin{aligned}\text{Required percentage is} &= \frac{280-180}{180} \times 100 \\ &= \frac{10000}{180} = \frac{500}{9} = 55\frac{5}{9} \% . \quad \text{Choice (D)}\end{aligned}$$

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.



In which month did the number of Xiaomi mobiles sold form the highest percentage of total mobiles sold in that month?

☐ a) March

☐ b) April

☒ c) May (Correct Answer - ✓)

☐ d) June

Solution:

The number of Xiaomi mobiles sold in the month of March = 180

Total mobiles sold in the month of March = 630

Required percentage, in

$$\text{March} \Rightarrow \frac{180}{630} \times 100 = 28.57\%$$

$$\text{April} \Rightarrow \frac{230}{860} \times 100 = 26.74\%$$

$$\text{May} \Rightarrow \frac{250}{800} \times 100 = 31.25\%$$

$$\text{June} \Rightarrow \frac{260}{920} \times 100 = 28.26\%$$

$$\text{July} \Rightarrow \frac{190}{790} \times 100 = 24.05\%$$

∴ In May, Xiaomi mobiles formed the highest percentage of total mobile sales.

Choice (C)

What is the total number of Moto - E mobiles sold during the given period?

☐ a) 1,30,000

☐ b) 1,24,000

☐ c) 80,000

☒ d) 13,000 (Correct Answer - ✓)

Solution:

The total number of Moto – E mobiles sold in all the given months

$$= (1500 + 2500 + 2800 + 3000 + 3200)$$

$$= 13,000.$$

Choice (D)

The number of Redmi mobiles sold in June decreased by what percentage from May?

☐ a) 22%

☒ b) $33\frac{1}{3}\%$ (Correct Answer - ✓)

☐ c) $42\frac{1}{3}\%$

☐ d) $49\frac{1}{6}\%$

Solution:

The number of Redmi mobiles sold in May = 1200

The number of Redmi mobiles sold in June = 800

$$\text{Required percentage} = \frac{1200 - 800}{1200} \times 100$$

$$= 33\frac{1}{3} \%$$

Choice (B)

The ratio of the total number of Moto - G mobiles sold in March and May to that of Xiaomi mobiles sold in April and June is the same as that of the number of Moto - G mobiles sold in August to the number of Xiaomi mobiles sold in August. If in August the number of Moto - G mobiles sold is 6000, then find the number of Xiaomi mobiles sold in August.

☐ a) 8,400 (Correct Answer - ✓)

☐ b) 4,950

☐ c) 7,200

☐ d) 9,500

Solution:

The total number of Moto - G mobiles sold in March and May together
= (2000 + 1500) = 3500.

The total number of Xiaomi mobiles sold in April and June together = (2300 + 2600)
= 4900.

$$\text{Ratio} = \frac{3,500}{4,900} = 5 : 7$$

Let 'x' be the number of Xiaomi mobiles sold in 2015.

$$\text{Given, } \frac{5}{7} = \frac{6000}{x}$$

$$\Rightarrow x = 1200 \times 7 = 8400.$$

Choice (A)

Which of the following can be concluded?

- a. During the given period of sales, the total number of mobile phones sold in any month increased every month over that of the previous month.
- b. During April, the number of Xiaomi mobiles sold was more than 12% of those sold in March.
- c. The total number of Moto - G mobiles sold is more than the number of Moto-E mobiles sold during the given period.

☐ a) Only (a)

☐ b) Only (c)

☒ c) Only (b) (Correct Answer - ✓)

☐ d) (a) and (b)

Solution:

Statement (a) is false, because from April to May the total sales of mobiles decreased.

From statement (b),

The number of Xiaomi mobiles sold in March = 180

The number of Xiaomi mobiles sold in April = 230

$$\text{Required percentage} = \frac{230 - 180}{180} \times 100$$

$$= 27.77\% \quad (\text{True})$$

From (c),

The total number of Moto - G mobiles sold in the given months = 1,010

The total number of Moto - E mobiles sold in the given months = 1,300.

(False)

Choice (C)